

Johnny Adrián García Betancourt

M.Sc. Student in Electrical Engineering • Control & Robotics, KFUPM

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Research Profile

Mechatronics engineer and M.Sc. researcher working at the intersection of Scientific Machine Learning (SciML), differential geometry, port-Hamiltonian systems, and physics-based modelling for nonlinear dynamical systems. My research integrates variational mechanics, passivity theory, differential flatness, and deep learning architectures to construct observer-free, structure-preserving representations directly from physical telemetry. Passionate about investigating how the internal geometry, symmetries, and energy invariants of neural representations interact with physical systems, with long-term research directions in scientific foundation models, structure-preserving learning, geometric deep learning, and inverse problems.

Education

M.Sc. Student in Electrical Engineering – Control & Robotics Track 2025–Present
King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia GPA: 3.938/4.00

Thesis: *Bridging Physical Safety and Active Tolerance in Robotic Manipulators: A Hierarchical Energy-Aware Architecture.*
Expected defense: December 2026–January 2027.

Selected Graduate Coursework: Linear Control Systems (A+), Optimal Control Theory & Applications (A+), Classical Mechanics (PHYS 507, A+), Advanced Methods for Theoretical Physics (PHYS 571, A+), Image Processing (A+), Special Topics in Intelligent Automation & Robotics (A+), Robotics & Control (A).

Graduate Coursework in Automation and Control (Master's Program) 2022–2024
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador Certified Coursework Avg: 9.97/10

Completed advanced graduate coursework with near-perfect marks; transferred to KFUPM prior to thesis completion.

Selected Coursework: Machine Learning-Based Control (100/100), Identification & Control of Dynamic Systems (100/100), Embedded Control Systems (100/100), Industrial Robotics & Computer Vision (97/100), Industrial Automation Systems (100/100).

B.Sc. in Mechatronics Engineering (Engineer in Mechatronics) 2017–2022
Escuela Superior Politécnica del Litoral (ESPOL), Ecuador Graduation Date: Dec 7, 2022

Thesis: *Mechanical and Electrical Design, and Automation of an Industrial Liquid Filling Machine* (Thesis Grade: 9.20/10).
Awarded Best Mechatronics Thesis.

Research Experience

Spectral-Hamiltonian Algebraic Power Evaluation (SHAPE Framework) Research Project | KFUPM

Conceived and formulated an observer-free, physics-guided fault detection and isolation methodology interfacing port-Hamiltonian energy balance ($\dot{H} \leq y^T u$) with differential flatness. Formulated the algebraic port-power residual that computes nominal power trajectories directly from reference profiles, eliminating online observer synthesis and numerical differentiation. Led the multi-body physical simulation and trajectory data-generation pipeline on a 6-DOF Kinova Gen3 manipulator. Transitioned the methodology to native numerical tensors (multichannel temporal, CWT time-frequency scalograms, and FFT spectral tensors), co-designed a shared-encoder multi-joint CNN with late fusion, and personally implemented the training, evaluation, and benchmarking pipeline in Python/PyTorch, achieving a 91.7% macro F1-score across nominal and degraded operating regimes.

Hierarchical Energy-Aware Architecture for Safe Robotic Interaction M.Sc. Thesis | KFUPM

Developing a hierarchical control framework bridging deterministic physical passivity with active fault tolerance in robotic manipulators. Formulated a low-level Port-Hamiltonian Energy Tank supervisor to enforce strict passivity ($\dot{H} \leq 0$) and bound energy injection under actuator Loss of Effectiveness (LoE). Developed a coordinate-free high-level trajectory planner parametrizing end-effector pose directly on the Lie group $SE(3)$ along with kinematic self-motion for 7-DOF redundant manipulators, ensuring dynamic feasibility without coordinate singularities.

Active Fault-Tolerant Flatness-Based Model Predictive Control Research Manuscript | Under Review

Developed an active fault-tolerant model predictive control architecture for nonlinear robotic manipulators subject to progressive actuator degradation. The formulation algebraically maps shrinking physical actuator torque limits into state-dependent acceleration constraints in flat output space, eliminating numerical chattering, reducing energy consumption by 66% (from 17,064 to 5,668 N²m²s), and accelerating trajectory optimization to real-time execution speeds (0.01 s).

Geometric Modelling and Invariant State Estimation Coursework & Research Projects | KFUPM

Investigated geometric mechanics and symmetry-preserving estimation for underactuated nonlinear dynamical systems. Developed multi-body dynamic models for an underactuated underwater vehicle actuated by internal shifting mass and variable-buoyancy piston systems. Implemented closed-loop simulation and Kalman filtering state estimation architectures in MATLAB, Simulink, and Simscape Multibody, analyzing state reconstruction accuracy under measurement noise and unmodeled hydrodynamics as a foundation for invariant and structure-preserving learning.

Research Manuscripts & Working Papers

[1] **J. Garcia Betancourt** and N. Salguero España, “Active Fault-Tolerant Flatness-Based MPC for Robotic Manipulators under Parametric Mismatch and Actuator Degradation,” *Research Manuscript*, under review, 2026.

[2] **J. Garcia Betancourt**, N. Salguero España, N. Iqbal, B. Brahmi, and R. Rashad, “Energy-Based Fault Detection and Isolation Framework for Flat Port-Hamiltonian Mechanical Systems,” *Research Manuscript*, in revision, 2026.

[3] U. I. Castilla Batun and **J. A. Garcia Betancourt**, “Techno-Economic Feasibility of Wind-Powered Hydrogen Production in Saudi Arabia: A Multi-Regional Assessment,” *Conference Paper*, under review, 2026.

Technical Expertise

Scientific ML & Deep Learning	Physics-guided representation learning; custom 1D/2D CNNs, shared-weight multi-joint encoders and late feature fusion; temporal, CWT time-frequency and FFT/spectral representations; comparative representation benchmarking; multi-seed evaluation, clustered bootstrap uncertainty quantification and robustness analysis; SVMs, PCA and t-SNE latent-space analysis.
Mathematical Foundations & Differential Geometry	Linear and tensor algebra, multivariable calculus, ordinary and partial differential equations, Fourier and Laplace methods, numerical methods, probability and statistics, calculus of variations, constrained and convex optimization; smooth manifolds, tangent and cotangent bundles, differential forms, Lie brackets, Riemannian metrics, Lie groups and Lie algebras ($SE(3)$, $SO(3)$, $\mathfrak{se}(3)$, $\mathfrak{so}(3)$), exponential maps, and differential-geometric modelling.
Analytical Mechanics & Energy-Based Physical Modelling	Variational, Lagrangian and Hamiltonian mechanics; symplectic and Poisson formulations, conservation laws and symmetries, rigid-body and multibody mechanics, twists, wrenches, adjoint/coadjoint transformations and spatial-vector mechanics; port-Hamiltonian systems, Dirac structures, passivity, energy shaping, energy tanks, and power/energy-based modelling.
Nonlinear Systems, Control & Optimization	Linear and nonlinear dynamical systems, state-space modelling, equilibrium and stability analysis, Lyapunov stability, LaSalle invariance, controllability and accessibility, geometric control, differential flatness, optimal control, MPC, Quadratic Programming (QP), trajectory optimization, inverse dynamics, and active fault-tolerant control.
State Estimation, Signal & Image Processing	Bayesian state estimation, Kalman filtering, Extended Kalman Filtering (EKF), Lie-group and invariant state-estimation concepts, SLAM foundations, Fourier and spectral analysis, digital filtering, Continuous Wavelet Transform (CWT), spatial and frequency-domain image processing, restoration, segmentation, mathematical morphology, feature extraction, pattern classification, and computer-vision foundations.
Scientific Computing & Engineering Tools	Python, PyTorch, MATLAB, Simulink, Simscape Multibody, Git, $\LaTeX$, CAD/CAE (SolidWorks, Inventor, ANSYS), instrumentation, PLC/SCADA systems, and industrial communication protocols.
Research interests: <i>scientific foundation models, structure-preserving learning, geometric deep learning, and inverse problems.</i>	

Academic and Professional Experience

Graduate Teaching & Laboratory Assistant <i>King Fahd University of Petroleum & Minerals (KFUPM), Saudi Arabia</i> Instructing undergraduate laboratory sessions in electronics and instrumentation, delivering hands-on guidance on circuit analysis, operational amplifiers, signal conditioning, and electronic measurement systems.	2025–Present
Teaching Assistant – Statistics and Probability <i>Escuela Superior Politécnica del Litoral (ESPOL), Ecuador</i> Assisted undergraduate engineering courses; conducted problem-solving recitations on probability distributions, parameter estimation, and statistical data analysis; evaluated student assignments.	2022
Undergraduate Research & Prototyping Assistant <i>Escuela Superior Politécnica del Litoral (ESPOL), Ecuador</i> Contributed to design and hardware prototyping for mechatronic machinery projects, assisting in CAD modeling, sensor interfacing, and PLC-based automated control routines.	2021–2022
Maintenance and Automation Coordinator <i>Detecuador S.A., Ecuador</i> Supervised automated operations, industrial instrumentation, and predictive maintenance protocols for manufacturing facilities; engineered 2D/3D CAD solutions for electro-mechanical system modernization.	2021–2024
Assistant Electromechanical Designer <i>SEOR S.A., Ecuador</i> Developed engineering blueprints, electrical control cabinets, and automation layouts for industrial machinery and packaging lines.	2023–2024

Honors and Awards

Graduate Full Scholarship , King Fahd University of Petroleum & Minerals (KFUPM)	2025–Present
Best Mechatronics Thesis Award (1st Place) , Escuela Superior Politécnica del Litoral	2022
Best University Thesis Award (2nd Place Institution-Wide) , ESPOL	2022
Academic Excellence Recognition (Top Score) , National University Admission Exam (SENESCYT)	2017

Languages

Spanish – Native	English – Advanced / Academic working proficiency
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